

Mediterranean Sea model validation draft

Field	Entry
Region	Mediterranean Sea, LME_026 Regional workbook
Catch source	Sea Around Us reconstructed catches, 1950–2019. Reference: 2019 landings, excluding discards. Frozen regional extract. Sea Around Us region Catch files
Selected article	Piroddi, Coll, Macias, Steenbeek, Garcia-Gorriz, Mannini, Vilas and Christensen (2022). Modelling the Mediterranean Sea ecosystem at high spatial resolution to inform the ecosystem-based management in the region. Scientific Reports 12, 19680. DOI Paper PDF Paper folder
Other known articles	Piroddi et al. (2017). Historical changes of the Mediterranean Sea ecosystem: modelling the role and impact of primary productivity and fisheries changes over time. Scientific Reports 7, 44491. 1950s baseline with 1950–2011 simulations and four sub-basin networks. 2017 article Local PDF
Selected model	Piroddi_2022_Mediterranean_1995. Basin-wide 1995 baseline (supplement: 1990s); 71 source groups: 65 consumers, 4 producers and 2 nonliving pools; 37 fleet entries. Model folder Selected JSON
Other models from the same article	One distinct Ecopath baseline is supplied. The 1995–2016 Ecosim/Ecospace simulations and spatial outputs are dynamic applications of that baseline, with no separately tabulated alternative Ecopath parameterization.
Selection rationale	Article — The recorded recommendation prefers the recent whole-basin source with supplied parameter-bearing XLSX and DOCX supplements. Model — Explicit user selection of the extracted 1995 model despite blocked construction; missing routing and the conditional fin-whale conflict remain.
Model extraction	Source B/PB/QB/EE, diets and fleet values are preserved. Diet sums including imports span 0.99956013582–1.00000013576; no normalization. GS, BA, migration and routing remain unknown. Diagnostic staging assumes zero unlisted landings but preserves reported discards; construction stops before requested GS/BA defaults. Fin-whale discards exceed production under the stated zero-BA/net-migration arithmetic; the residual is unexplained and unapplied. Source audit Lossless roundtrip
SPPR calculation	[MANUAL — Researcher: calculation choices and reasons] [Excluded groups, scope and relevant values; other adjustments]

Field	Entry
GE diagnostics	NOT_RUN: missing routing between Discards (70) and Detritus (71) blocks construction. Negative SPPR entries cannot be assessed. rho_living and b are unavailable; SPPR for Discards (70) and Detritus (71) is unavailable.
TE diagnostics	NOT_RUN: the same constructor restriction prevents calculation. Negative SPPR entries and rho_living cannot be assessed; SPPR for Discards (70) and Detritus (71) is unavailable.
Geographic fit	A. Region covered by study: approximately 95–100%. B. Study area covered by region: approximately 95–100%.
Temporal fit	The 1995 baseline precedes reference catch year 2019 by 24 years. Fixed source composition and stage proxies are transferred across 1950–2019 catches and bases. These do not reconstruct annual ecosystem structure; model PPR remains unavailable.
Other	With Egestion also remains NOT_RUN. Unknown routing cannot be replaced by a source balance check or a taxonomic mapping. Constructor and method evidence
Open issues and next action	[Researcher: open issues and next actions]
Review and reproducibility	Researcher name: [name] review date: [date]

Taxon mapping and coverage

Reference: 2019 landings. Independent all-taxon simple-chain estimate; all source support, all recorded labels and no group filter.

71 source groups; 522 catch taxa. No synthetic computational import group.

Simple-chain PPR totals 105,846,498.898 t C in 2019.

40 taxa lack a classic TL/coefficient; all have zero 2019 landings and contribute zero annual PPR. No positive-catch contribution is unknown.

Simple-chain PPR uses the fixed taxon coefficients of the simple trophic chain method.

[All-taxon Excel appendix](#) | [Descriptive Sources sheet](#)

Overall confidence	Taxa (n)	Catch (%)	Simple-chain PPR (%)
High	290	41.3944%	64.9733%
Medium	186	48.2426%	31.1116%
Low	26	0.1836%	0.1535%
Very low	20	10.1794%	3.7615%
Unresolved	0	0.0000%	0.0000%

The summary uses the weaker required membership or allocation confidence. The following tables show each component separately.

Group assignment rules

Membership evidence: [Source group definitions and authoritative taxonomic audit](#) Scaridae denotes the historical parrotfish family, now Scarinae within Labridae; the genus-misspelling homonym is excluded. [Scaridae rank bridge](#)

Plain-language rule	Confidence	PPR percentage
Explicitly listed source member of the selected group or stage pair	High	73.4641%
Assumed eligible group set supported by source members and taxonomy	Medium	14.9221%
Supported ecological extension of the source-defined group	Medium	6.7841%
Broad reporting category approximated by named and residual source pools	Very low	3.7615%
Taxonomic extension from source-listed representatives	Medium	0.3737%
Unambiguous fit to a documented taxonomic group with no competing pool	High	0.2710%
Verified synonym of an explicitly listed source member	High	0.2699%
Partial or conflicting ecological fit among plausible source pools	Low	0.1535%
Closest represented taxonomic or ecological analogue for an absent compartment	Very low	<0.0001%

Allocation weight rules

Weights and assumptions: [All candidates and source quantities](#) | [Primary-source composition search](#)

Plain-language rule	Confidence	PPR percentage
One group receives the complete taxon catch; no split uncertainty	High	79.2924%
Selected-model caught-mass proportions used as a fixed composition proxy	Medium	16.3223%
Complete accepted model-biomass proportions when source landings are unusable	Medium	4.3853%

Each rule percentage is the share of independent simple-chain PPR associated with taxa using that rule.

Unresolved taxa: none. Very low approximations account for 10.1794% of landings and 3.7615% of simple-chain PPR. They concern broad mixed categories and absent-compartment analogues; exact taxa, assumptions and weights are flagged in the appendix. Residual fish reporting uses bony-fish candidates; dedicated shark/ray pools are excluded. Mixed NC large pelagic retains *Mola mola* with cartilaginous pooling uncertainty. Complete source catch or biomass supplies assumed weights; model methods remain NOT_RUN. [Reporting scope evidence](#) The retained crustacean proxy includes documented decapods^{53–60}, mixed nondecapod benthos⁶¹ and unspecified zooplankton⁶⁵. The latter supplies 76.34% of the biomass split; this is unobserved composition, not caught-mass evidence. Broad aquatic-invertebrate pools remain a weak ecological proxy; provider demersal metadata is not exhaustive taxonomy.

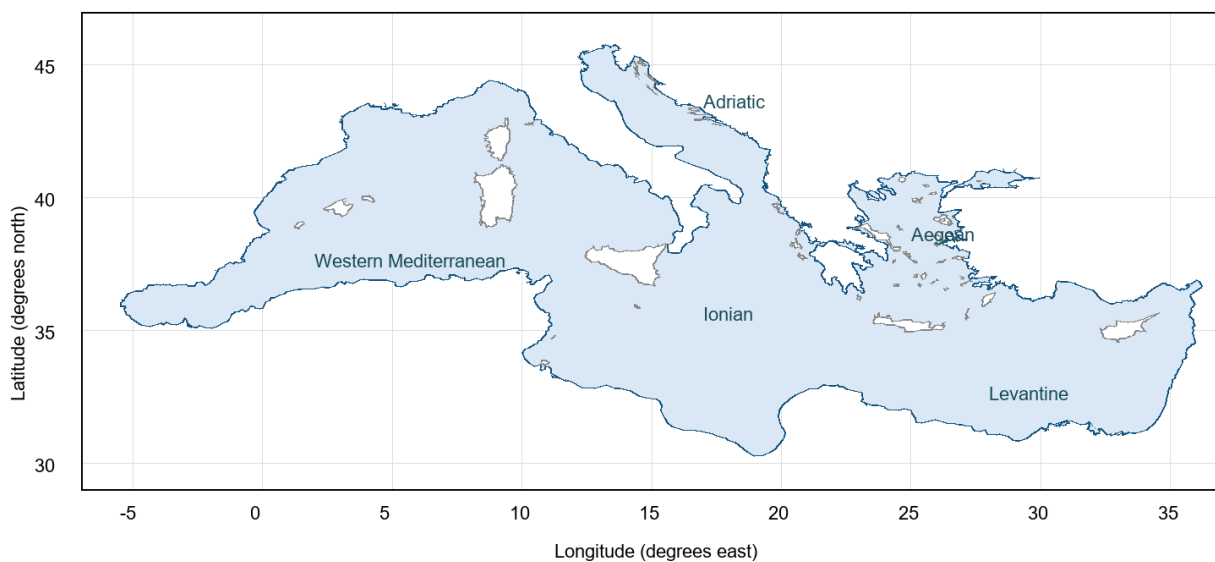
Geographic evidence screenshots

Target region R and selected-model study area S. The basin-wide geographic estimate is approximate; coastal grid cells and strait boundaries remain uncertain.

Region area

Target region boundary

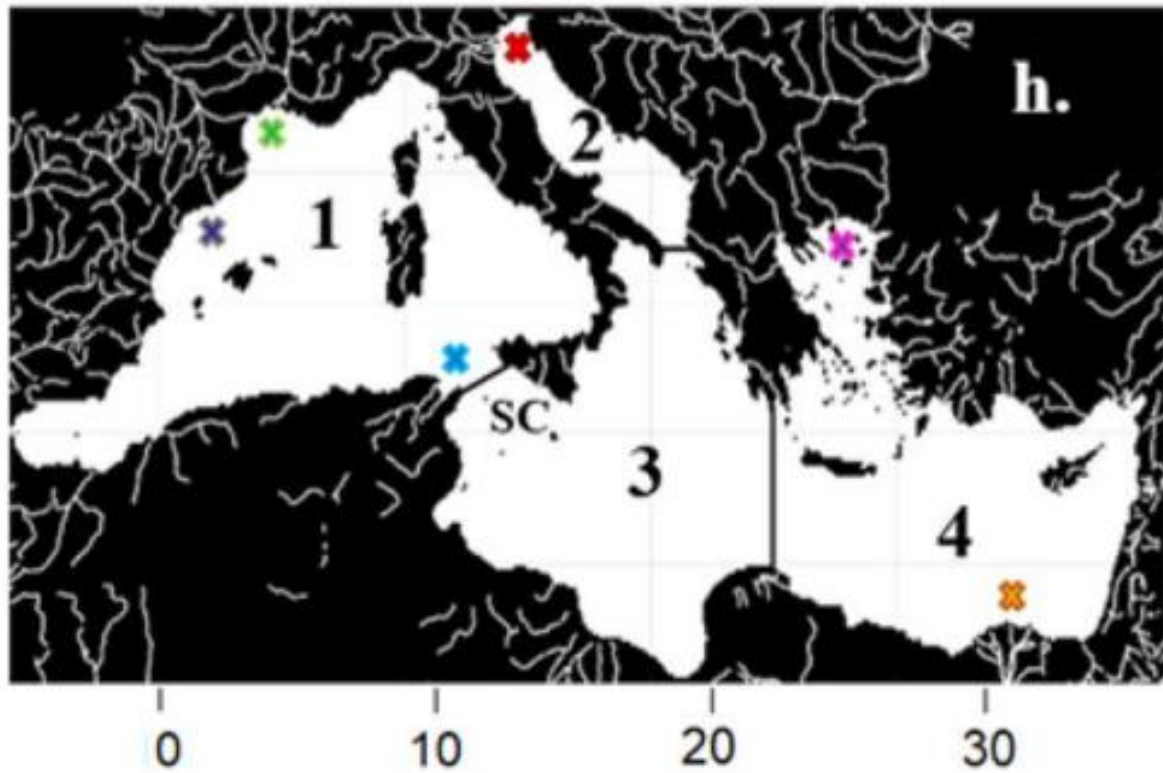
Mediterranean Sea LME 026 target marine boundary



Mediterranean Sea LME_026. Current project marine polygon. [Boundary source](#) | [Geographic assessment](#)

Study area in the article

Study boundary for the selected model



Piroddi et al. (2022), Figure 4h, p. 9. Whole Mediterranean ocean domain and four sub-basins; Figure 4a provides the matching marine mask. [Source PDF](#)